

DESIGN AND ANALYSIS OF ALGORITHM (CS-327)

LECTURE – 5

RECURRENCE RELATION :-

A recurrence relation is an equation which represents a sequence based on some rule. Four types are given below.

1. Substitution method
2. Iteration method
3. Recursive tree method
4. Master method

SUBSTITUTION METHOD :-

(i) Forward substitution:-

Forward substitution is a method which uses the initial condition in initial term and generate the next term and keep on repeating the same process until we guess particular formula. We rarely use this method because we will have to guess the formula in this method.

Example :-

$$T(n) = T(n-1) + n, T(0)=0$$

$$T(0)=0 \quad (\text{initial condition})$$

Solution :-

Let $n=1$

$$T(1)=T(0)+1$$

$$T(1)= 1\text{-----eq(i)}$$

Let $n=2$

$$T(2)= T(1) + 2$$

$$T(2) = 1 + 2$$

$$T(2) = 3 \text{ -----eq(ii)}$$

Let $n=3$

$$T(3) = T(2) + 3$$

$$T(3) = 3 + 3$$

$$T(3) = 6 \text{ -----eq(iii)}$$

$$T(n) = \frac{n(n+1)}{2}$$

$$T(n) = n^2/2 + n/2$$

$$T(n) = O(n^2).$$

Backward substitution method:-

In this method backward values are substituted recursively. We generally use Backward substitution.

Example :-

$$T(n) = T(n-1) + n ; T(0) = 0$$

$$T(0) = 0 \text{ initial condition}$$

Solution :-

$$T(n) = T(n-1) + n \text{ -----eq(i)}$$

So we will start from back value, so replace n with $n-1$ in eq(i)

$$T(n-1) = T(n-2) + (n-1) \text{ -----eq(ii)}$$

Now we will replace n with $n-2$

$$T(n-2) = T(n-2-1) + (n-2)$$

$$T(n-2) = T(n-3) + (n-2) \text{ -----eq(iii)}$$

$$T(n-k) + (n-k+1) + (n-k+2) + \dots \dots \dots n$$

If $k=n$ then

$$T(n) = T(0) + 1 + 2 + 3 + 4 + 5 + \dots + n$$

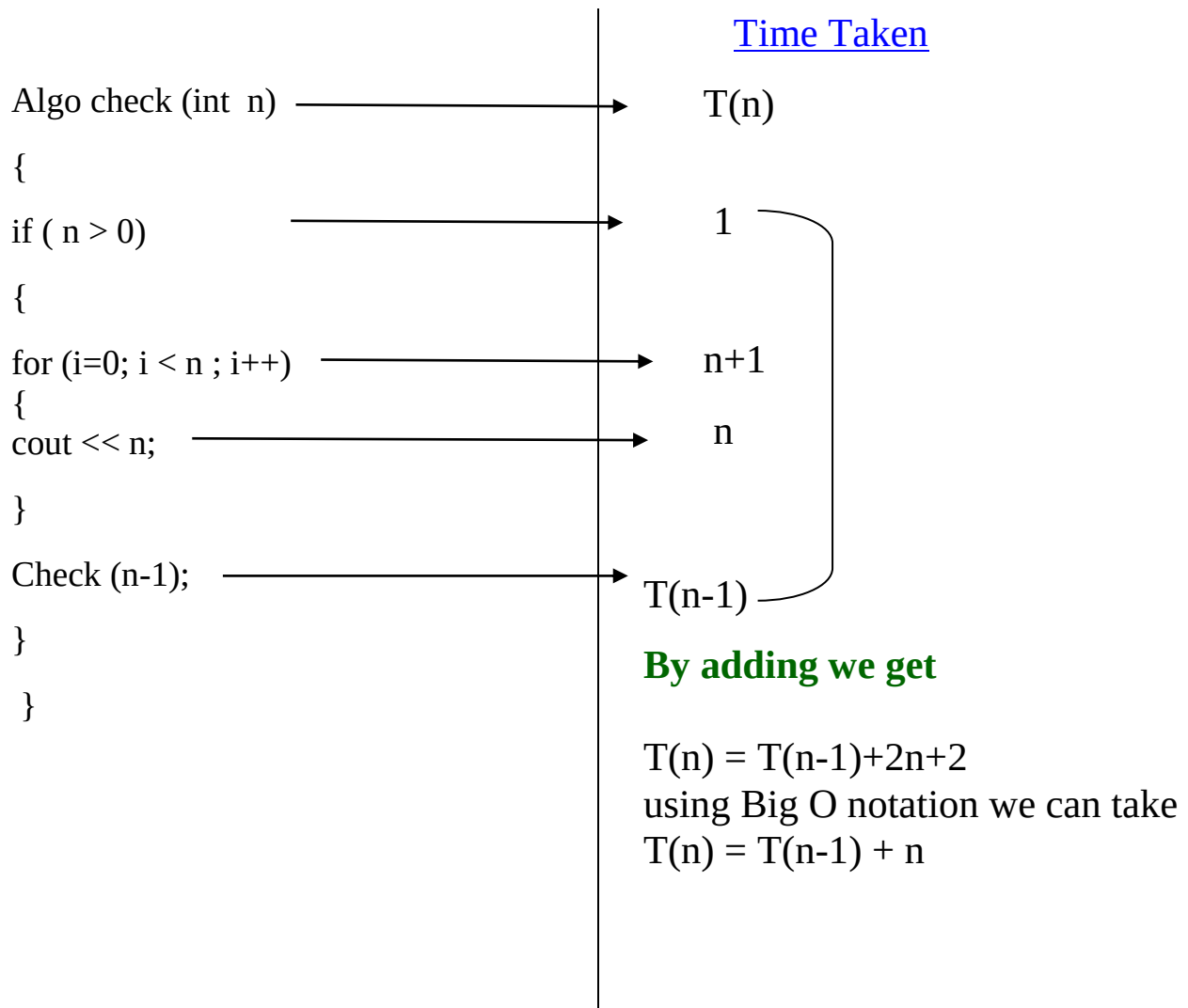
$T(n)$ = sum of natural numbers

$$T(n) = \frac{n(n+1)}{2}$$

$$T(n) = n^2/2 + n/2$$

$$T(n) = O(n^2)$$

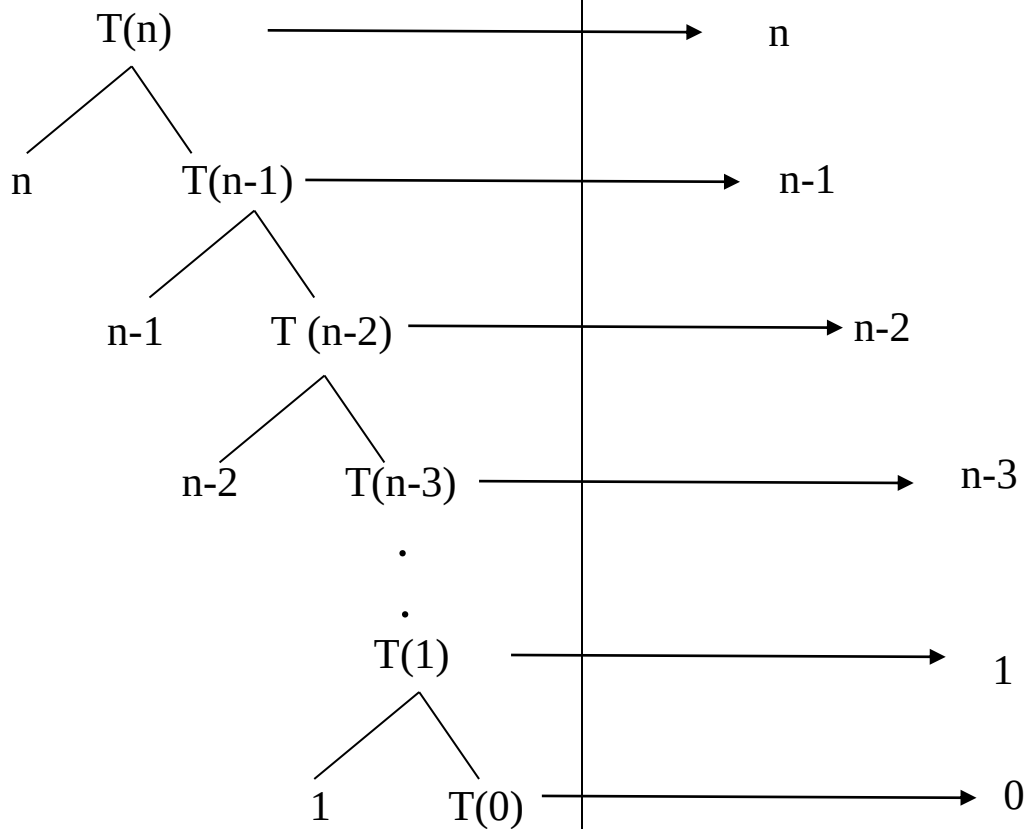
RECURSIVE TREE METHOD:-



$$T(n) = T(n-1) + n$$

$$T(n) = \begin{cases} 1 & n = 0 \\ T(n-1) + n & n > 0 \end{cases}$$

Lets solve recurrence tree of the above recurrence relation.



So by adding all above time we get

$$0 + 1 + 2 + n - 1 + n + \dots = \frac{n(n+1)}{2}$$

$$T_n = \frac{n^2}{2} + \frac{n}{2}$$

$$\mathbf{T(n) = O(n^2).}$$